Google Cloud

Course 3 resources and citations

Module 1: Introduction to data transformation in the cloud

Resources

Helpful resources and tips

- Common problems with labs/lab troubleshooting
- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

The data journey is a flexible path

- 7 data collection methods in business analytics

Impacts of the data journey on decision making

- Business analytics in action: Real-world examples and use cases

Techniques Used to Transform Data, Part 2

- Importance of proper formatting in data collection

Storage options for transformed data

- BigQuery overview

Citations

Stages of the data journey

- Statistics Canada. (2021, October 22). *Data journey*.

The data journey is a flexible path

- Cymetrix. (2023, January 11). *Business analytics in action: Real-world examples and use cases*. Linkedin.

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- Weinmeister, K. (2020, April 13). From raw data to machine learning model, no coding required. *Google Cloud Blog*.

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Steps for effective data collection

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- Cote, C. (2021, December 2). 7 data collection methods in business analytics. *Business Insights Blog*.

The process stage ensures clean and consistent data

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- Choudhury, A. (2021, January 22). *Top 8 data transformation methods*. Analytics India Magazine.
- Linkedin. (n.d.). *What are some common data transformation techniques and when to use them?*

[Supplemental] Process and summarize data

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- The University of Utah. (n.d.). *Central tendency & variability*.

Techniques Used to Transform Data, Part 2

- Hamilton, T. (2020, March 31). What are Data Unions? How do they work? Which ones can I use? *Streamr*.
- W3 schools. (n.d.). *SQL joins*.

Manual and automated transformation

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- Google Cloud. (n.d.). *What is ETL?* Retrieved July 20, 2023.

Storage options for transformed data

- Enagandula, R. (2023, May 17). *Database vs data warehouse vs data lake*. Linkedin.

Module 2: Handle raw data with data pipelines

Resources

Overview of data pipeline stages and benefits

- Data Pipelines: An Overview
- What Data Pipeline Architecture should I use?

Strategies and tools for efficient data ingestion

- Big data ingestion

Data transformation steps to ensure reliable data

- What are the most common data cleaning techniques?

Transformation approaches for securing data

- FBI Tech Tuesday: Protecting Against Personally Identifiable Information (PII) Theft
- How to safeguard personally identifiable information

Data dictionaries and monitoring help maintain data integrity

- Definitive Guide to Data Monitoring
- What Is a Data Dictionary?

Analysis and visualization after the data pipeline

- How to Meet WCAG (Quick Reference)
- Introduction to SQL

- Top data visualization tools for business professionals

Citations

Data pipelines in the cloud

- Dowsett, C. (2022, December 22). *What is a data pipeline?* Built In.
- Google Cloud. (n.d.). *What is ETL?* Retrieved July 20, 2023.

Overview of data pipeline stages and benefits

- Knight, M. (2023, March 2). Data pipelines: An overview. *DATAVERSITY*.

The difference between ELT and ETL

- Google Cloud. (n.d.). *What is ETL?* Retrieved July 20, 2023.
- Jameel, M. Y. (2022, August 26). *ETL vs ELT?* Linkedin.

Data ingestion methods that suit your needs

- Gomede, E. (2023, August 2). Batch vs. streaming data ingestion: Choosing the right approach for efficient data processing. *Medium*.
- Rajan, V. (2023, July 25). *What is data ingestion and why it is important*. Linkedin.
- Thallam, R. (2020, September 16). BigQuery explained: How to ingest data into BigQuery so you can analyze it. *Google Cloud Blog*.

Strategies and tools for efficient data ingestion

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Data mapping helps ensure consistent data

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Data transformation steps to ensure reliable data

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Introduction to profiling and cleaning data

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- Google for Developers. (2022, July 18). *Representation: Cleaning data*. Machine Learning. Retrieved July 25, 2023.
- Indeed Editorial Team. (2022, October 12). *What is data profiling? (Definition, techniques and types)*. Indeed.

Common ways to manipulate data

- CESSDA ERIC. (n.d.). *File formats and data conversion*. Data Management Expert Guide.
- Kline, L. (2021, October 7). Understanding data enrichment: A comprehensive guide | Hightouch. *The Data Activation Blog*.
- Linkedin. (n.d.). *What are the benefits and challenges of data standardization and normalization?*
- Linkedin. (2024, January 13). *What are the best tools and methods for data conversion and loading?*

Data transformation techniques that achieve different business needs

- Kulshrestha, A. (Ace), & Ellis, S. (2019, June 26). Take charge of your data: How tokenization makes data usable without sacrificing privacy. *Google Cloud Blog*.
- Linkedin. (n.d.). *What are the benefits and challenges of data standardization and normalization?*
- Linkedin. (2023, December 8). *What are some effective strategies for data deduplication in the ETL process?*

Transformation approaches for securing data

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Different approaches to loading data

- Chetry, N. (2023, August 12). Incremental data load approach and implementation strategy. *Medium*.

- Google Cloud. (2024, January 12). *Introduction to loading data*.

Overview of data validation strategies and rules

- Metwalli, S. A. (2023, March 7). *What is data validation?* Built In.
- Nene, N., & Fu, J. (2021, July 21). Introducing the data validation tool for EDW migrations. *Google Cloud Blog*.

Data dictionaries and monitoring can help you maintain data integrity

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- UC Merced Library. (n.d.). *What is a data dictionary?*

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Module 3: Cloud data optimization strategies

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Analyze a data transformation plan

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Data aggregation can make it easier to extract meaningful insights

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Consequences of duplicate data and how to eliminate it

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